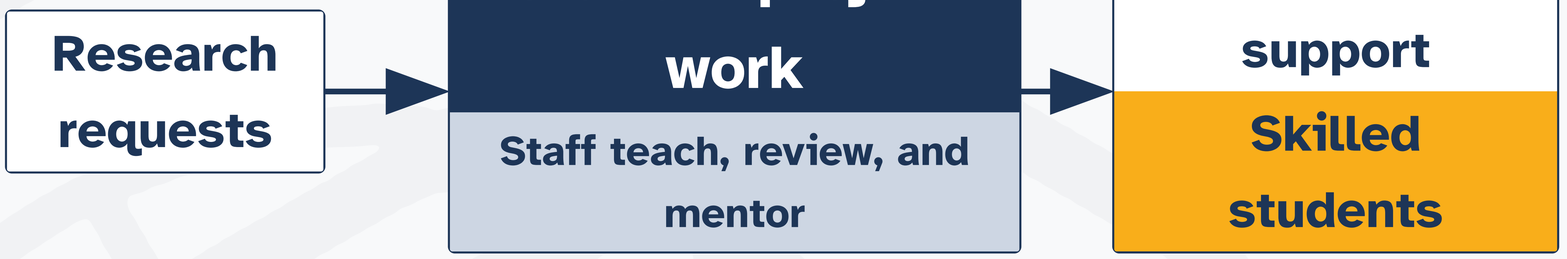
DataSquad: An entry point for RSE

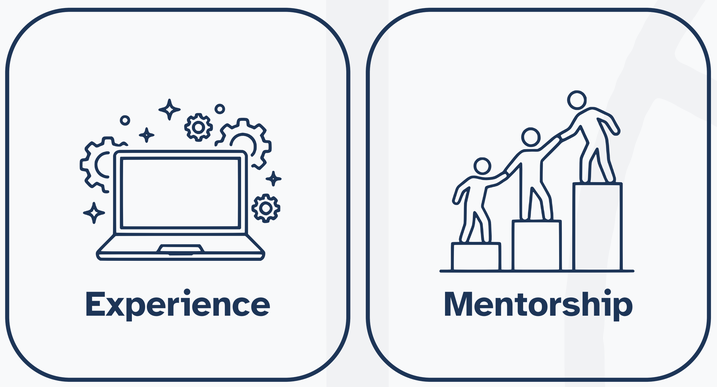
Student RSE model for research support, experiential learning, and technical workforce development

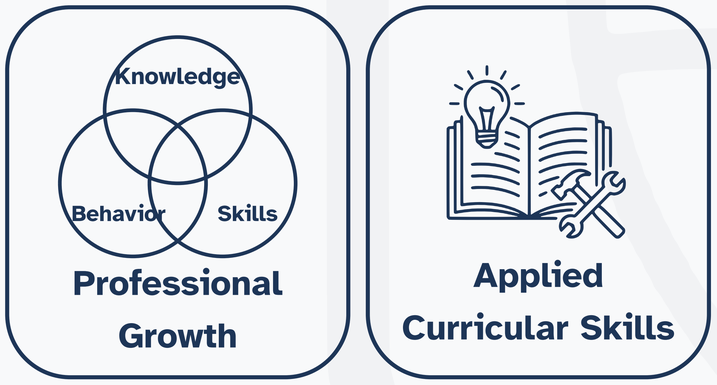
Paula Lackie, Oliver Hall, Tim Dennis, Leigh Phan
Carleton College DataSquad | UCLA Library Data Science Center DataSquad

Paid students contribute to research and build RSE skills while they do it.

Campuses need more data and software help than staff can provide. Courses don't cover everything, and the code researchers write often goes without support. DataSquads close that gap.







DataSquad roles

Project Management Intern
Triages requests, helps manage and mentor their peers, advises the staff

Data Analysts and Developers
Do the hands-on work with data and code

Technical Writers
Document projects and assure clear communications.

Staff
Supervise, mentor, and situate the DataSquad in their institution

What the work looks like

CARLETON Election-news scraper. Seven students built it in 2022. Two more extended it from 2023 to 2024 for a political science project that involved over 500,000 transcripts.



Each dot is one student. Blue built it, orange extended it.

UCLA Police helicopter flight data. Two students brought software skills to a research lab's R code. They refactored it with staff, and the code went through code review. Loading the data fell from about 318 minutes to about 2.



LAPD helicopter, for illustration. Photo: Chris Yarzab, CC BY 2.0, via Wikimedia Commons. Cropped and resized.

Time to load the data



Research software habits, when the project needs them. Run instructions, a dependency list, one test, and a citation file. We do not treat them as a checklist. We plan to track whether researchers keep them. Two student-built projects have citation files and DOIs that credit their developers: wbcheck (**doi:10.5281/zenodo.23329531**) and the UC OSPO Education site (**doi:10.5281/zenodo.23330022**).

Where students go next

Where former student employees work now, in the words of their job titles.



Words in the job titles of 96 alumni (UCLA and Carleton), from public profiles in 2026. Larger means more people; words used by fewer than 3 people are not shown. Navy is technology, steel is data and research, amber is other. Descriptive, not a placement rate.

What adopting takes

- Student positions and a wage budget
- Supervisors with time to review and coach
- Onboarding and a home for documentation
- A plan for turnover

Start a DataSquad

Get the starter kit for roles, onboarding, confidentiality, and request tracking. Students learn software sharing from our open lesson, **Sharing Research Software**. Supervise students? Compare onboarding, review, and graduation handoffs with us.

datasquad.info/join

doi:10.5281/zenodo.23381638